

SOC Day 1

Introduction to Security Operations Center and Log Analysis

1. Introduction to SOC

SOC stands for Security Operations Center. A Security Operations Center is a team that monitors systems, networks, websites, endpoints, and servers to detect cyber attacks and suspicious activity. SOC teams are responsible for protecting organizations from attackers. They continuously monitor systems because attacks can happen at any time. SOC teams are commonly used in: Banks Government organizations IT companies Cloud providers Military organizations Large enterprises The main purpose of SOC is to detect attacks quickly and reduce damage before attackers gain full control over systems.

2. Main Responsibilities of a SOC Team

SOC teams perform many important tasks every day. Main responsibilities include: Monitoring alerts SOC analysts continuously monitor alerts generated by firewalls, SIEM tools, antivirus systems, IDS, IPS, and endpoint security tools. Log analysis Logs help analysts understand what happened inside systems. Threat detection SOC teams identify suspicious activity like brute force attacks, malware execution, privilege escalation, and unusual network traffic. Incident response When an attack is detected, SOC analysts investigate and help stop the attack. Threat hunting Analysts manually search for hidden attackers that may not generate alerts. Reporting Every investigation is documented properly for future analysis and compliance.

3. SOC Team Structure

A SOC team usually contains multiple roles. Tier 1 Analyst Tier 1 analysts monitor alerts and perform initial investigation. Their work includes: Checking suspicious login attempts Monitoring dashboards Reviewing alerts Escalating serious incidents Tier 2 Analyst Tier 2 analysts perform deeper investigations. They analyze: Attacker behavior Malware activity Privilege escalation Persistence

mechanisms Tier 3 Analyst Tier 3 analysts investigate advanced attacks and perform threat hunting. They also: Create detection rules Improve monitoring Analyze advanced malware Build investigation workflows Incident Response Team This team handles active attacks and helps recover systems after incidents.

4. Understanding Logs

Logs are records of activities happening inside systems. Every action inside a computer or server generates logs. Examples: User login attempts Website requests File access Command execution Network connections USB device insertion System errors Logs are very important because attackers always leave traces behind. SOC analysts use logs to: Detect attacks Investigate incidents Build timelines Identify attacker IP addresses Understand attacker behavior Without logs, investigation becomes extremely difficult.

5. Important Linux Log Files

Linux systems contain many important log files. `/var/log/auth.log` Stores: SSH login attempts Successful logins Failed passwords sudo activity `/var/log/syslog` Stores general system events and service activity. `/var/log/apache2/access.log` Stores website traffic and HTTP requests. `/var/log/apache2/error.log` Stores website errors and suspicious attack attempts. `/var/log/wtmp` Stores successful login history. `/var/log/btmp` Stores failed login attempts. SOC analysts frequently investigate these logs during incidents.

6. Understanding a Real Log Entry

Example log: May 11 14:22:11 kali sshd[2201]: Failed password for root from 192.168.1.20 port 4455 ssh2 This log contains important information. Timestamp Shows when the event happened. Hostname Shows the system generating logs. Process Shows which process generated the event. Event type Shows what happened. Target account Shows which account was targeted. Source IP Shows attacker IP address. SOC analysts quickly extract these details during investigations.

7. Common Attack Patterns

SOC analysts search for suspicious patterns inside logs. Brute force attack Many failed password attempts from the same IP. Password spray attack Few password attempts against many users. Privilege escalation A normal user suddenly gains root access. Persistence New hidden accounts are

created. Suspicious login Login occurs at unusual times. Malware activity Unexpected processes start running. These patterns help analysts identify attacks quickly.

8. Important SOC Commands

SOC analysts use Linux commands daily during investigations. View failed logins: `grep 'Failed password' /var/log/auth.log` View successful logins: `grep 'Accepted password' /var/log/auth.log` Monitor logs live: `tail -f /var/log/auth.log` View login history: `last` View failed login history: `lastb` View open ports: `netstat -tulnp` View running processes: `ps aux` These commands are heavily used during investigations.

9. What is SIEM

SIEM means Security Information and Event Management. A SIEM platform collects logs from multiple systems into one centralized dashboard. Popular SIEM tools include: Splunk ELK Stack Wazuh QRadar Graylog Benefits of SIEM: Centralized monitoring Real time alerts Faster investigations Better visibility Dashboard reporting Large organizations use SIEM systems because manually checking logs from thousands of systems is difficult.

10. Threat Hunting

Threat hunting means actively searching for attackers inside systems. Instead of waiting for alerts, analysts manually investigate suspicious activity. Threat hunters search for: Malware Hidden users Suspicious PowerShell commands Data exfiltration Lateral movement Persistence mechanisms Threat hunting requires patience, observation, and analytical thinking.

11. Incident Response Process

Incident response means handling attacks properly and systematically. Main steps: Detection Attack is identified. Analysis Evidence is collected and analyzed. Containment The attacker is stopped from spreading. Eradication Malware and persistence are removed. Recovery Systems are restored safely. Reporting Investigation findings are documented. SOC teams follow these steps during real incidents.

12. Practical Investigation Example

Suppose an attacker performs SSH brute force attacks. Investigation process: Check failed logins in auth.log. Identify attacker IP address. Check if login became successful. Review commands executed by attacker. Check if new users were created. Investigate suspicious network connections. Create investigation timeline. Block attacker access. This is how real SOC investigations happen.

13. Skills Required for SOC

To become a strong SOC analyst, students should learn: Linux Networking Log analysis Basic scripting SIEM tools Incident response Threat detection Students should also improve: Observation skills Patience Documentation habits Analytical thinking

14. Final Notes

SOC is one of the best domains for students entering cybersecurity. SOC teaches: Investigation Defensive security Log analysis Threat detection Incident handling A good SOC analyst always follows evidence carefully and continuously improves investigation skills.

Quick Command Reference

Purpose	Command
Failed logins	grep 'Failed password' auth.log
Successful logins	grep 'Accepted password' auth.log
Live monitoring	tail -f auth.log
Login history	last
Failed login history	lastb
Running processes	ps aux
Open ports	netstat -tulnp